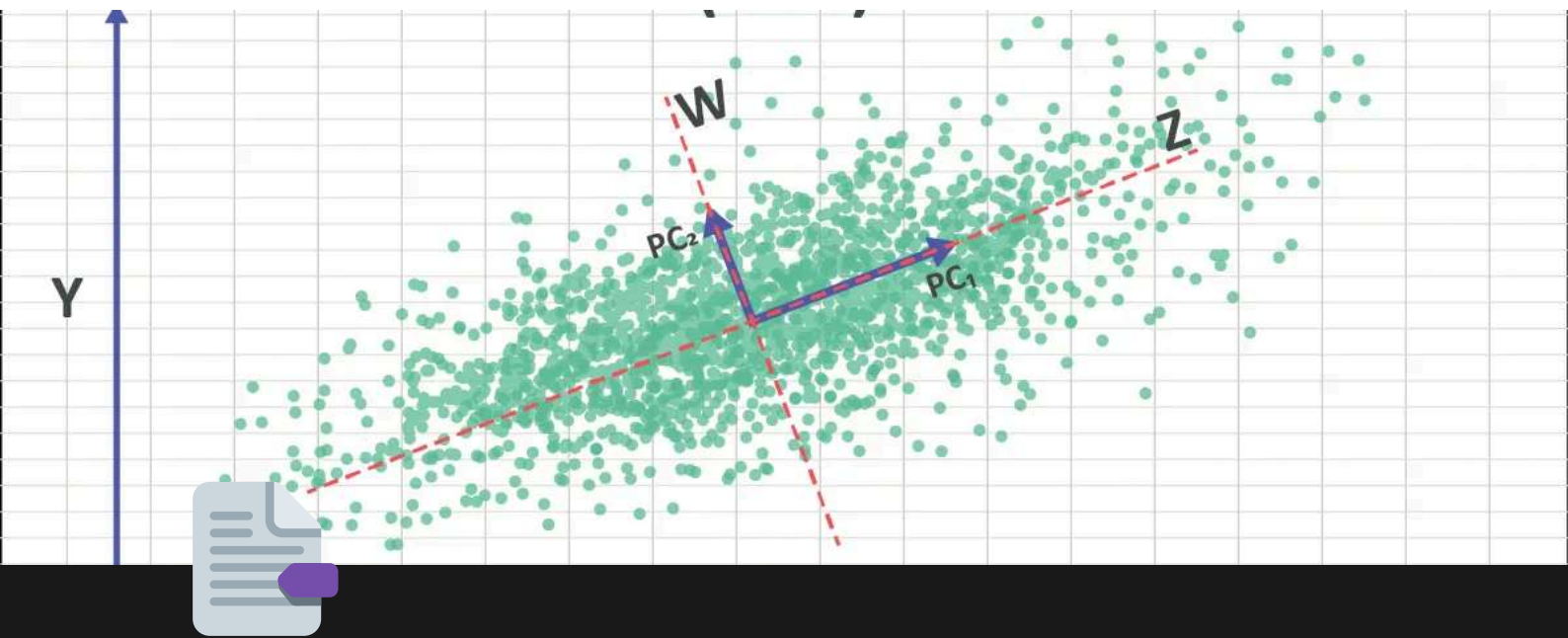




PCA - Principal Component Analysis

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Date: 16th Sept 2025

What is PCA?

Principal Component Analysis (PCA) is a dimensionality reduction technique. It transforms high-dimensional data into a smaller set of uncorrelated variables called **principal components**, while preserving as much variance (information) as possible.

Think of it as:

- You have data in many dimensions (say, 10 features).
- PCA finds new "axes" (directions) in the data that capture the **most variance**.
- Then it re-expresses the data using those fewer axes.

Why PCA?

- **Remove redundancy:** If features are correlated, PCA reduces them into fewer uncorrelated components.
- **Visualization:** Reduce data to 2D or 3D for plotting.

- **Speed:** Algorithms run faster on fewer features.
 - **Noise reduction:** Small-variance components are often just noise.
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Steps of PCA

1. Standardize the data

Since PCA is variance-based, features with large scales dominate.

→ Convert each feature to mean = 0 and variance = 1.

2. Compute covariance matrix

Measures how features vary together.

$$\text{Cov}(X) = \frac{1}{n-1} (X^T X)$$

(where X is mean-centered data).

3. Find eigenvalues and eigenvectors

- Eigenvectors = principal components (directions of maximum variance).
- Eigenvalues = amount of variance captured by each component.

4. Sort eigenvalues (and corresponding eigenvectors)

Highest eigenvalue = 1st principal component, and so on.

5. Project data

Multiply original data by selected eigenvectors to reduce dimensions.

Numerical Example

Suppose we have a dataset with 2 features:

Person	Height (cm)	Weight (kg)
A	170	65
B	180	72
C	160	60
D	175	70
E	165	63

Step 1: Standardize

Compute mean & subtract.

- Mean height = 170, mean weight = 66.
- Centered data:

Person	Height-mean	Weight-mean
A	0	-1
B	10	6
C	-10	-6
D	5	4
E	-5	-3

Step 2: Covariance matrix

For two variables x and y , covariance is:

$$\text{Cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

We do this for:

$$\text{Var}(x) = \text{Cov}(x, x)$$

$$\text{Var}(y) = \text{Cov}(y, y)$$

$$\text{Cov}(x, y)$$

Step 2.1: Variance of Height

$$\begin{aligned}\text{Var}(\text{height}) &= \frac{1}{4}(0^2 + 10^2 + (-10)^2 + 5^2 + (-5)^2) \\ &= \frac{1}{4}(0 + 100 + 100 + 25 + 25) = \frac{250}{4} = 62.5\end{aligned}$$

Step 2.2: Variance of Weight

$$\begin{aligned}\text{Var}(\text{weight}) &= \frac{1}{4}((-1)^2 + 6^2 + (-6)^2 + 4^2 + (-3)^2) \\ &= \frac{1}{4}(1 + 36 + 36 + 16 + 9) = \frac{98}{4} = 24.5\end{aligned}$$

Step 2.3: Covariance (Height, Weight)

$$\begin{aligned}\text{Cov}(\text{height}, \text{weight}) &= \frac{1}{4}(0 \cdot -1 + 10 \cdot 6 + (-10)(-6) + 5 \cdot 4 + \\ &\quad (-5)(-3)) \\ &= \frac{1}{4}(0 + 60 + 60 + 20 + 15) = \frac{155}{4} = 38.75\end{aligned}$$

Step 2.4: Assemble the covariance matrix

$$\text{Cov} = \begin{bmatrix} 62.5 & 38.75 \\ 38.75 & 24.5 \end{bmatrix}$$

Step 3: Eigenvalues & eigenvectors

- Eigenvalues $\approx 83.0, 2.0$.

- Eigenvectors (normalized):
 - $PC1 \approx [0.85, 0.53]$
 - $PC2 \approx [-0.53, 0.85]$

Step 3: Eigenvalues and Eigenvectors

We now solve the **eigenvalue equation**:

$$\text{Cov} \cdot v = \lambda v$$

where

- v is the eigenvector (the direction),
 - λ is the eigenvalue (variance captured in that direction).
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Step 3.1: Characteristic equation

We find eigenvalues by solving:

$$\det(\text{Cov} - \lambda I) = 0$$

That is:

$$\det \left(\begin{bmatrix} 62.5 - \lambda & 38.75 \\ 38.75 & 24.5 - \lambda \end{bmatrix} \right) = 0$$

Expanding determinant:

$$(62.5 - \lambda)(24.5 - \lambda) - (38.75)(38.75) = 0$$

$$= (1531.25 - 87\lambda + \lambda^2) - 1501.56$$

$$= \lambda^2 - 87\lambda + 29.69 = 0$$

So the eigenvalues are:

$$\lambda_1 = 86.66, \quad \lambda_2 = 0.34$$

Step 3.2: Eigenvectors

For $\lambda_1 = 86.66$:

Solve $(\text{Cov} - 86.66I)v = 0$:

$$\begin{bmatrix} 62.5 - 86.66 & 38.75 \\ 38.75 & 24.5 - 86.66 \end{bmatrix} = \begin{bmatrix} -24.16 & 38.75 \\ 38.75 & -62.16 \end{bmatrix}$$

The equations are:

$$-24.16x + 38.75y = 0$$

$$38.75x - 62.16y = 0$$

Both simplifies to the same ratio:

$$y = 0.6234x$$

So Eigen vector = $[1, 0.6234]$

Normalize (so length = 1):

$$v_1 = \left[\frac{1}{\sqrt{1^2 + 0.6234^2}}, \frac{0.6234}{\sqrt{1^2 + 0.6234^2}} \right] = [0.849, 0.529]$$

This is the **first principal component (PC1)**.

Step 4: Choose top components

- PC1 explains ~99.61% of variance $\frac{86.66}{86.66+0.34}$.
- PC2 explains only ~0.39%.

So we can reduce 2D $\rightarrow$ 1D without losing much information.

Step 5: Project data

Person	Height-mean	Weight-mean
A	0	-1
B	10	6
C	-10	-6
D	5	4
E	-5	-3

Take dot product of centred data with PC1:

Person	Projection (PC1)
A	-0.529
B	11.664
C	-11.664
D	6.361
E	-5.832

Now the dataset is **1D** but still captures almost all variance.

Real-World Example